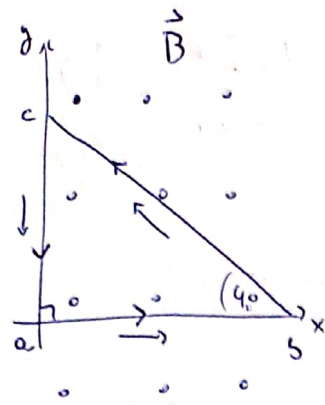




$$l_{ac} = 0,18 \text{ m}$$

$$\sin 40 = \frac{l_{ac}}{l_{bc}}$$

$$l_{bc} = 0,28 \text{ m}$$

$$\tan 40 = \frac{l_{ac}}{l_{bs}}$$

$$l_{bs} = 0,215 \text{ m}$$

$$\Rightarrow \vec{l}_{ac} = -0,18 \hat{j} \quad (1)$$

$$\vec{l}_{bs} = 0,215 \hat{i} \quad (1)$$

$$\vec{B} = 0,500 \hat{k} \text{ [T]} \quad \vec{l}_{bc} = 0,28 \cos 40 (-\hat{i}) + 0,28 \sin 40 \hat{j}$$

$$= -0,215 \hat{i} + 0,18 \hat{j} \quad (2)$$

$$\Rightarrow \vec{F}_{ac} = I \vec{l}_{ac} \times \vec{B}$$

$$= 1,87 \cdot (-0,18 \hat{j}) \times 0,500 \hat{k}$$

$$= 0,1683 (-\hat{i}) \text{ [N]} \quad (5)$$

$$\vec{F}_{bs} = I \vec{l}_{bs} \times \vec{B}$$

$$= 1,87 \cdot 0,215 \hat{i} \times 0,500 \hat{k}$$

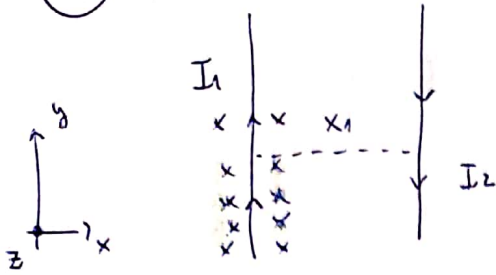
$$= 0,2010 (-\hat{j}) \text{ [N]} \quad (5)$$

$$\vec{F}_{bc} = I \vec{l}_{bc} \times \vec{B}$$

$$= 1,87 \cdot (-0,215 \hat{i} + 0,18 \hat{j}) \times 0,500 \hat{k}$$

$$= 0,2010 \hat{j} + 0,1683 \hat{i} \text{ [N]}. \quad (5)$$

2-



$$a) \vec{F}_1 = I_1 \vec{l}_1 \times \vec{B}_2$$

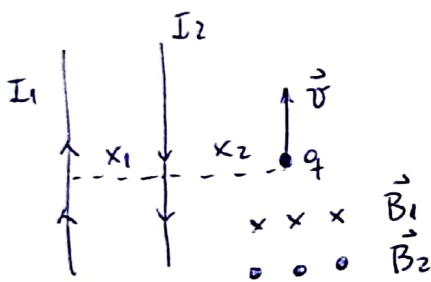
$$= I_1 l \hat{j} \times \frac{\mu_0 I_2}{2\pi x_1} (-\hat{u}) \quad (3)$$

$$= \frac{\mu_0 I_1 I_2 l}{2\pi x_1} (-\hat{u})$$

$$\frac{\vec{F}_1}{l} = \frac{4\pi \cdot 10^{-7} \cdot 0,500 \cdot 1,600}{2\pi \cdot 0,13} (-\hat{u})$$

$$(4) = 1,231 \cdot 10^{-6} (-\hat{u}) \text{ [N/m]}$$

b)



$$\vec{F} = q \vec{v} \times \vec{B} \quad \vec{v} = 250 \hat{j} \text{ [m/s]}$$

$$\vec{B} = \vec{B}_1 + \vec{B}_2$$

$$= \frac{\mu_0 I_1}{2\pi(x_1+x_2)} (-\hat{u}) + \frac{\mu_0 I_2}{2\pi x_2} \hat{u} \quad (3)$$

$$= \frac{\mu_0}{2\pi} \left(\frac{I_1}{(x_1+x_2)} (-\hat{u}) + \frac{I_2}{x_2} \hat{u} \right)$$

$$= \frac{4\pi \cdot 10^{-7}}{2\pi} \left(\frac{0,5}{0,2} (-\hat{u}) + \frac{1,6}{0,07} \hat{u} \right)$$

$$= 2 \cdot 10^{-7} (2,5 (-\hat{u}) + 22,86 \hat{u})$$

$$(2) = 4,072 \cdot 10^{-6} \hat{u} \text{ [T]}$$

$$\Rightarrow \vec{F} = 76 \cdot 10^{-9} \cdot 250 \hat{j} \times 4,072 \cdot 10^{-6} \hat{k}$$

$$= 7,737 \cdot 10^{-11} \hat{i} \text{ [N]}. \textcircled{5}$$

3.

$$B = \mu_0 n I \quad \textcircled{2} : \text{Campo del solenoide.}$$

a)

$$\phi = \int_R \vec{B} \cdot d\vec{S}$$

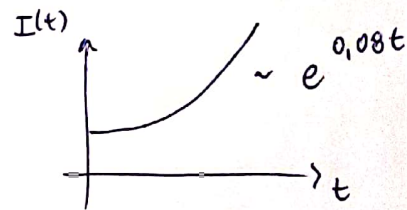
: Flujo a través de la bobina (1 vuelta)

$$= \int_R B dS$$

$$= B \int_R dS$$

$$= B (\pi R^2)$$

3



$$\Rightarrow \mathcal{E} = -N \frac{d\phi}{dt}$$

$$N = 15$$

$$= -N \mu_0 n \pi R^2 \frac{dI}{dt} \quad \textcircled{2}$$

$$= -\mu_0 N n \pi R^2 \frac{d}{dt} (0,4 e^{0,08t})$$

$$= -4\pi \cdot 10^{-7} \cdot 15 \cdot 220 \cdot \pi \cdot (0,075)^2 \cdot 0,4 \cdot 0,08 \cdot e^{0,08t}$$

$$= -2,345 \cdot 10^{-6} e^{0,08t} \text{ [V]}. \textcircled{3}$$

$$b) I = \frac{|\mathcal{E}|}{R}$$

$$= 1,117 \cdot 10^{-7} e^{0,08t} \text{ [A]}. \textcircled{5}$$

Dado que I aumenta en " t ", el flujo y el campo magnético también aumentan en el tiempo. Por tanto, la corriente inducida tiene la dirección de circulación opuesta a la del solenoide, de manera de evitar cambios en el flujo.

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